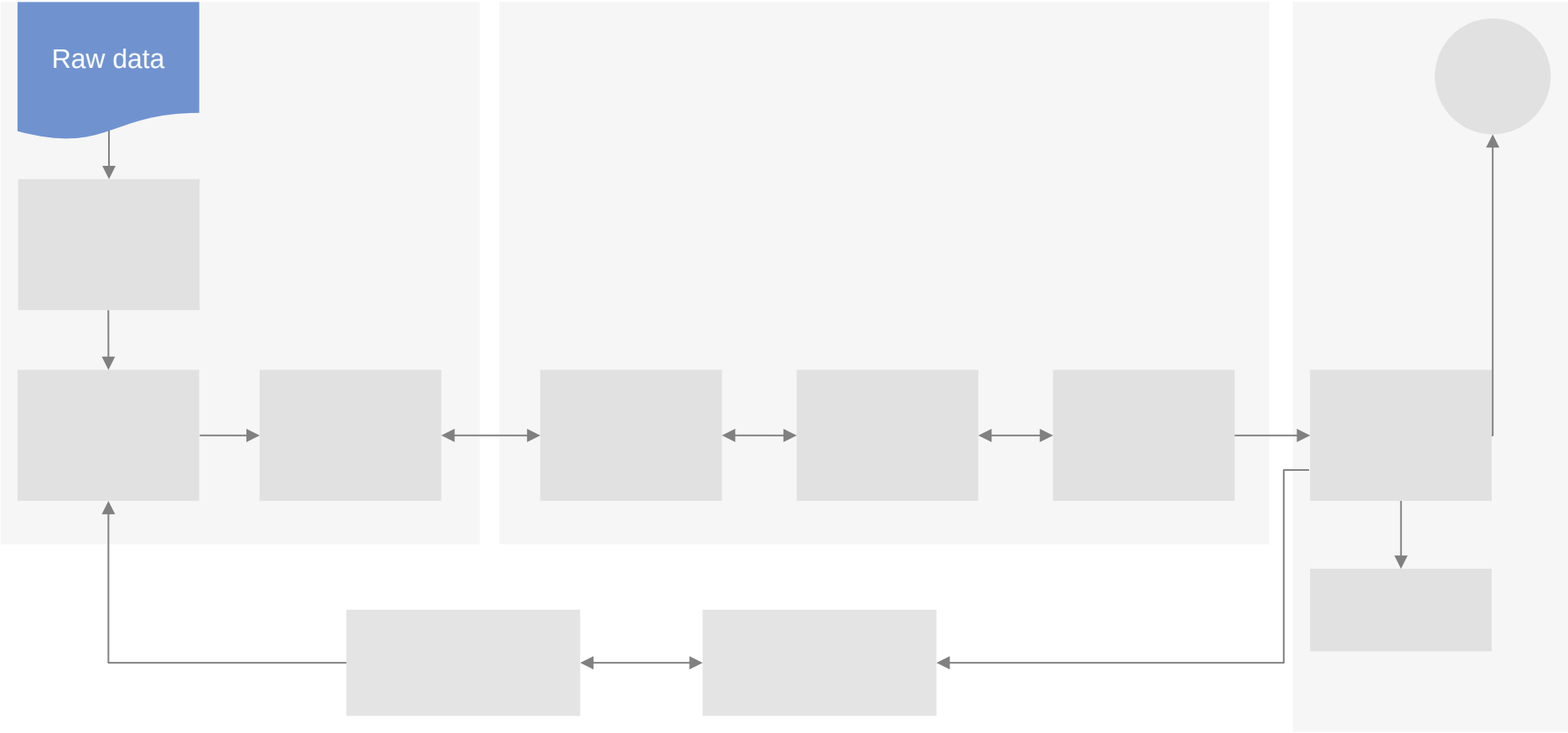
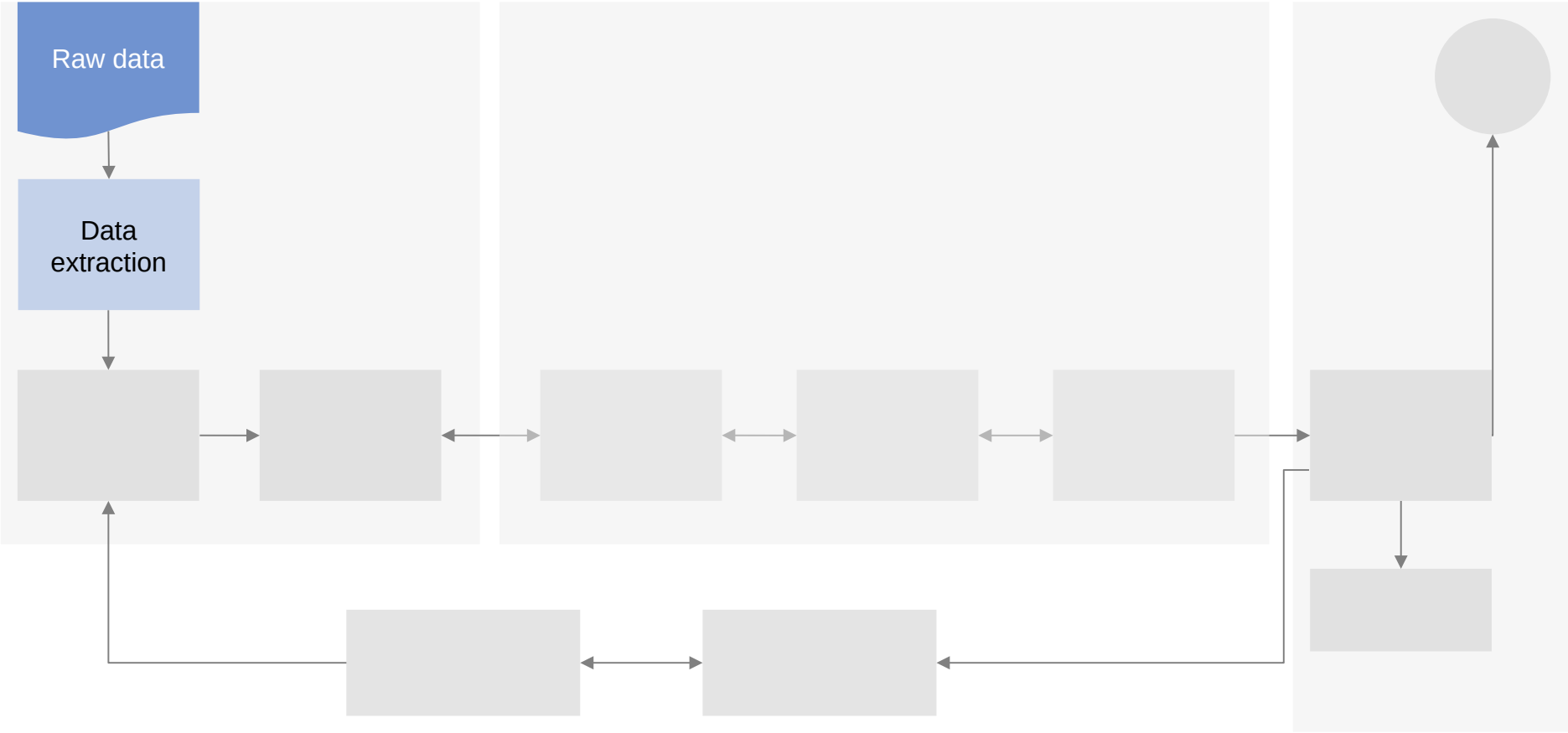
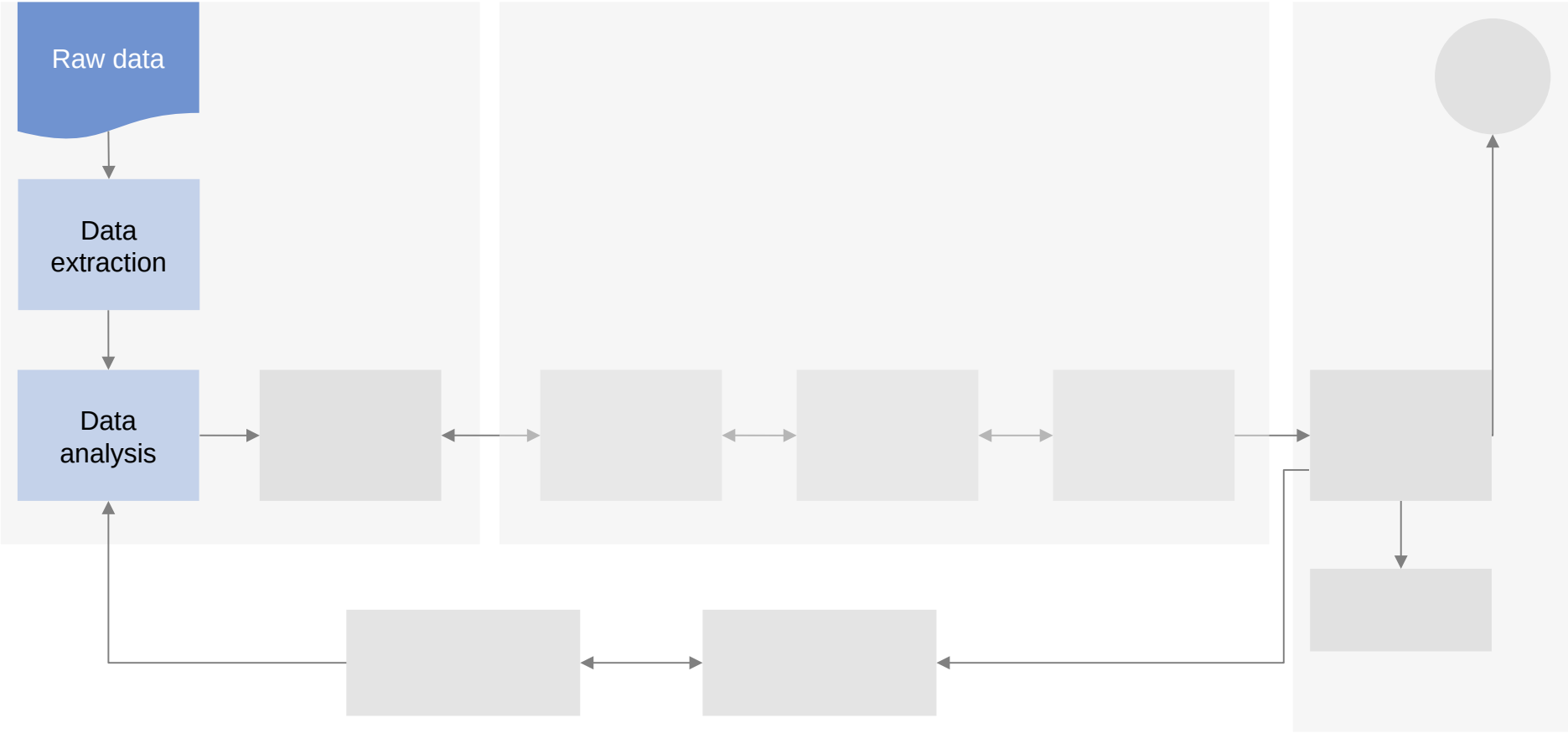
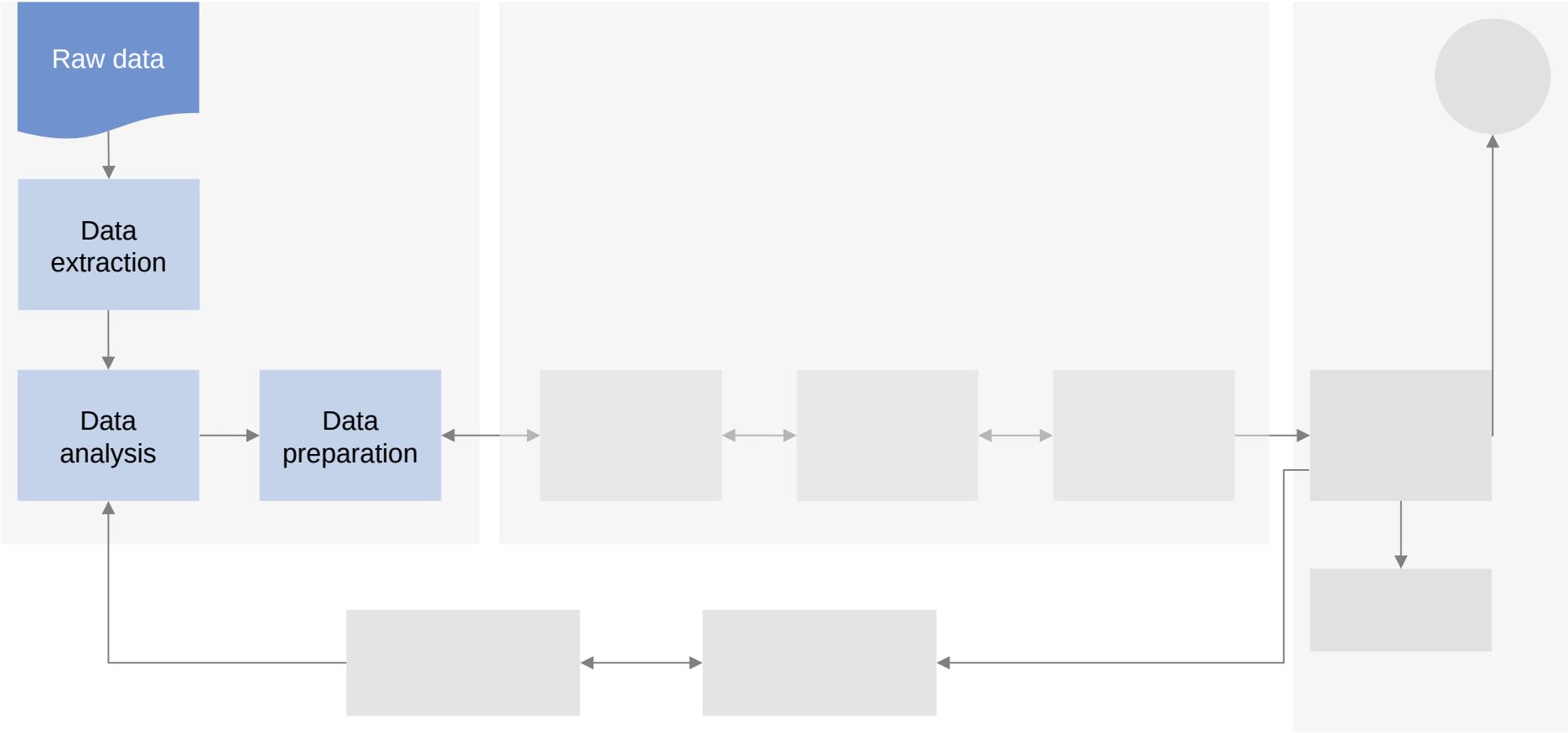
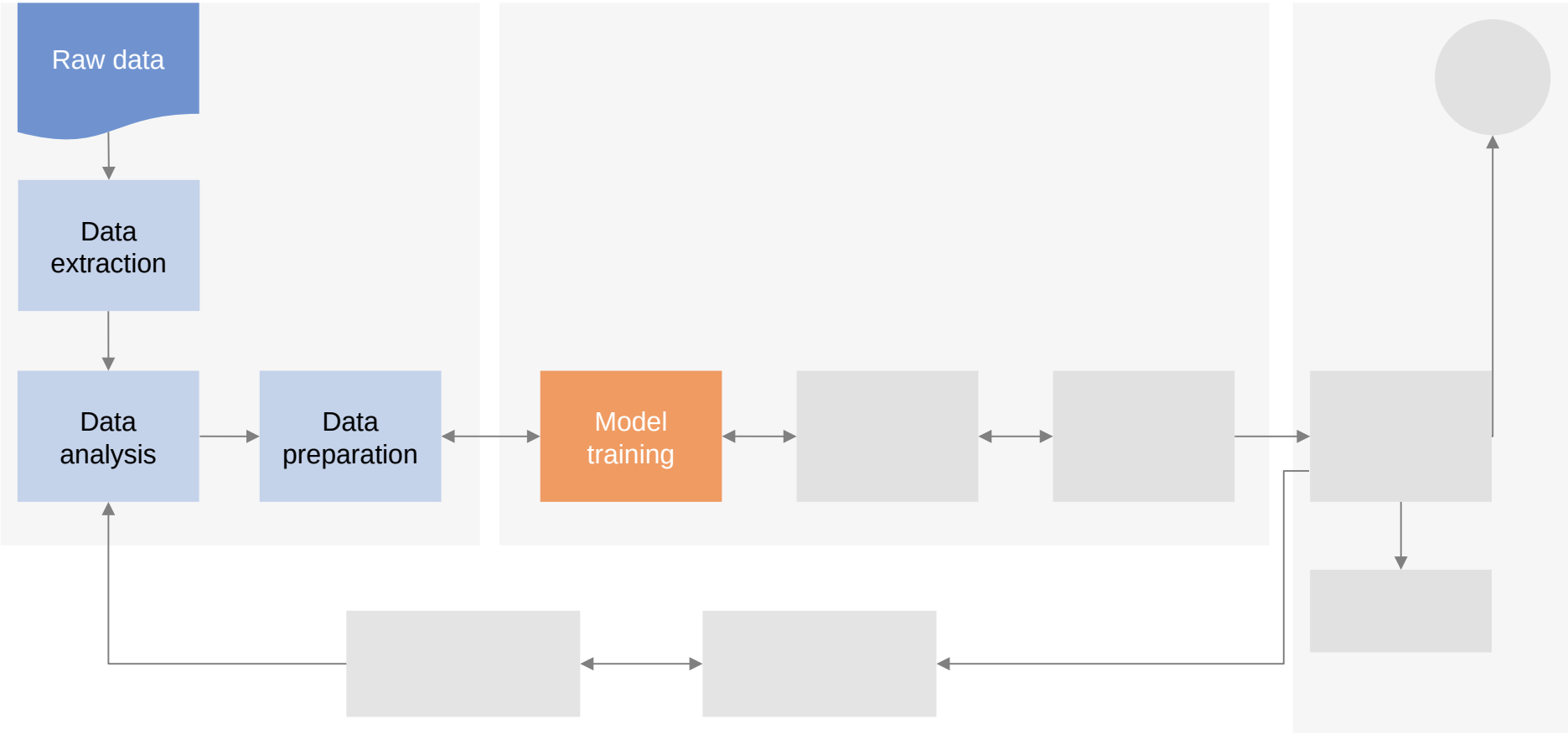
ML Model Lifecycle

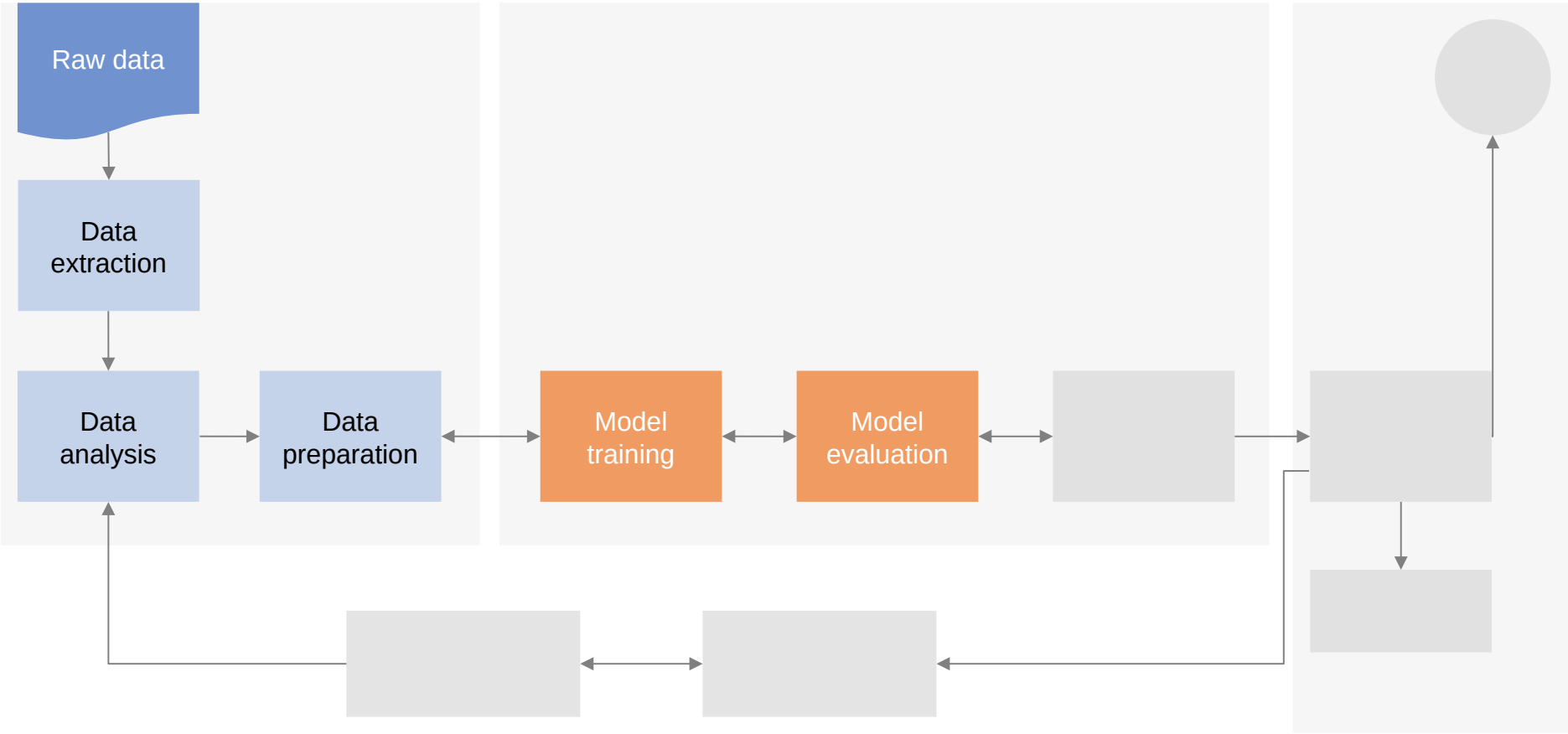

Raw data

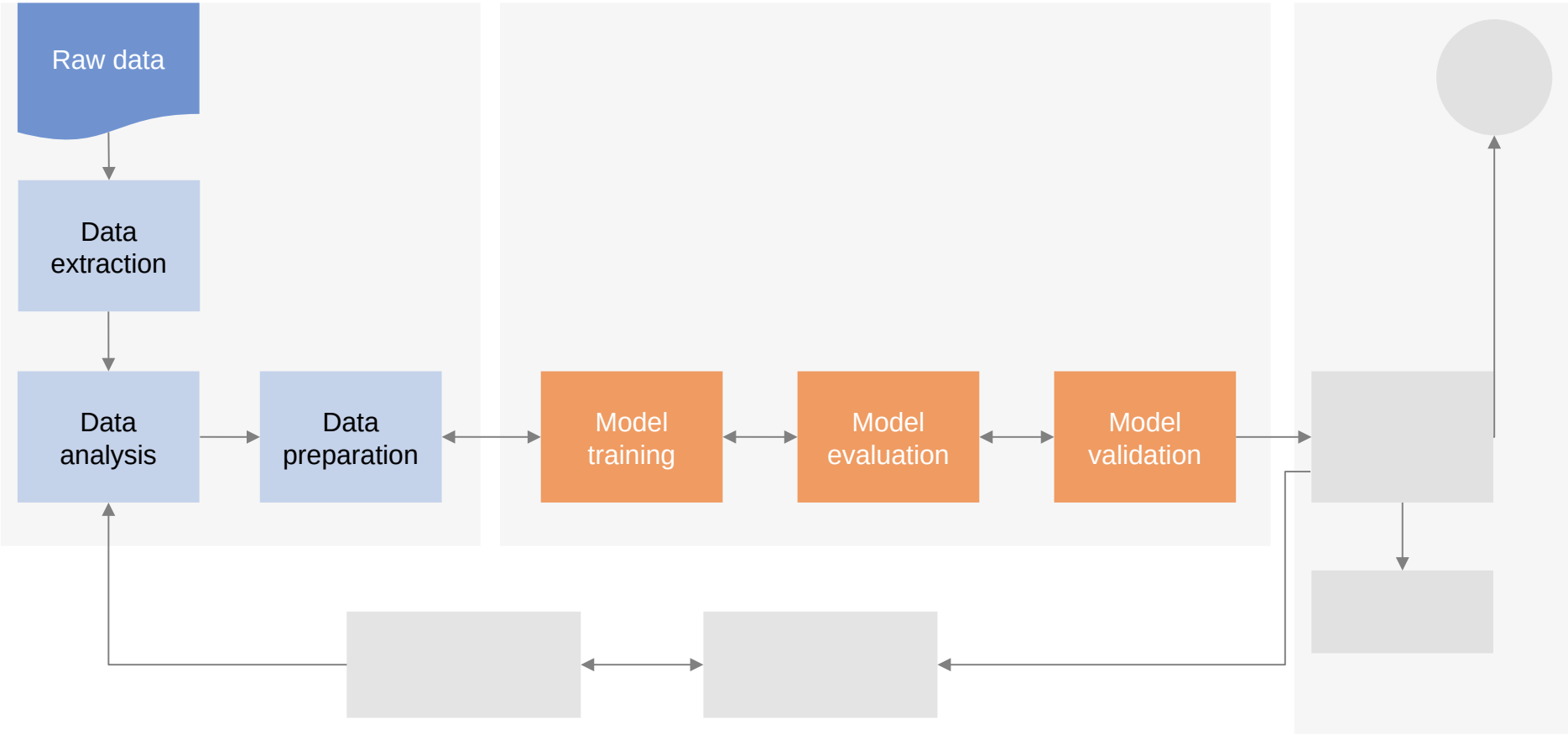

Raw data
Data extraction

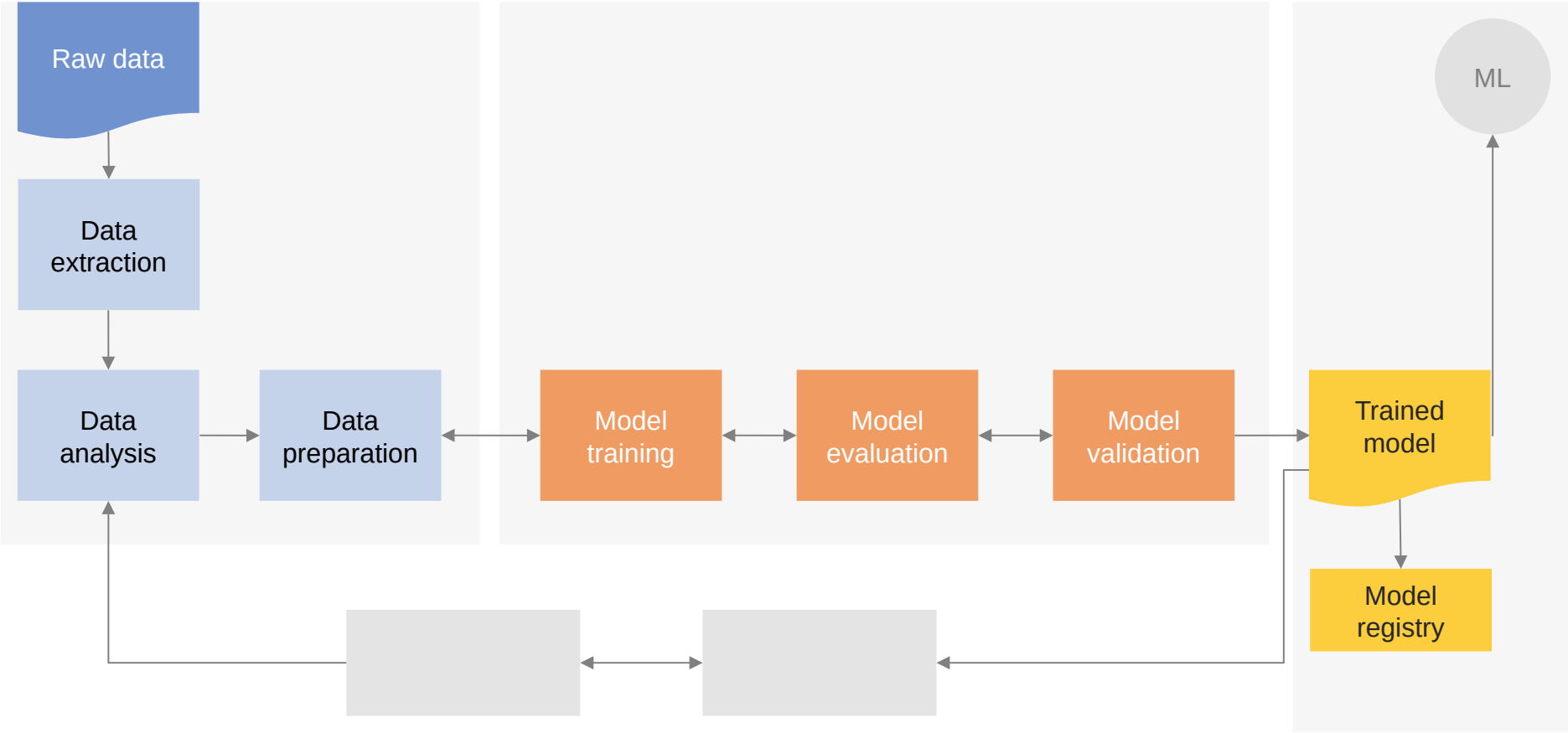

Raw data
Data extraction
Data analysis

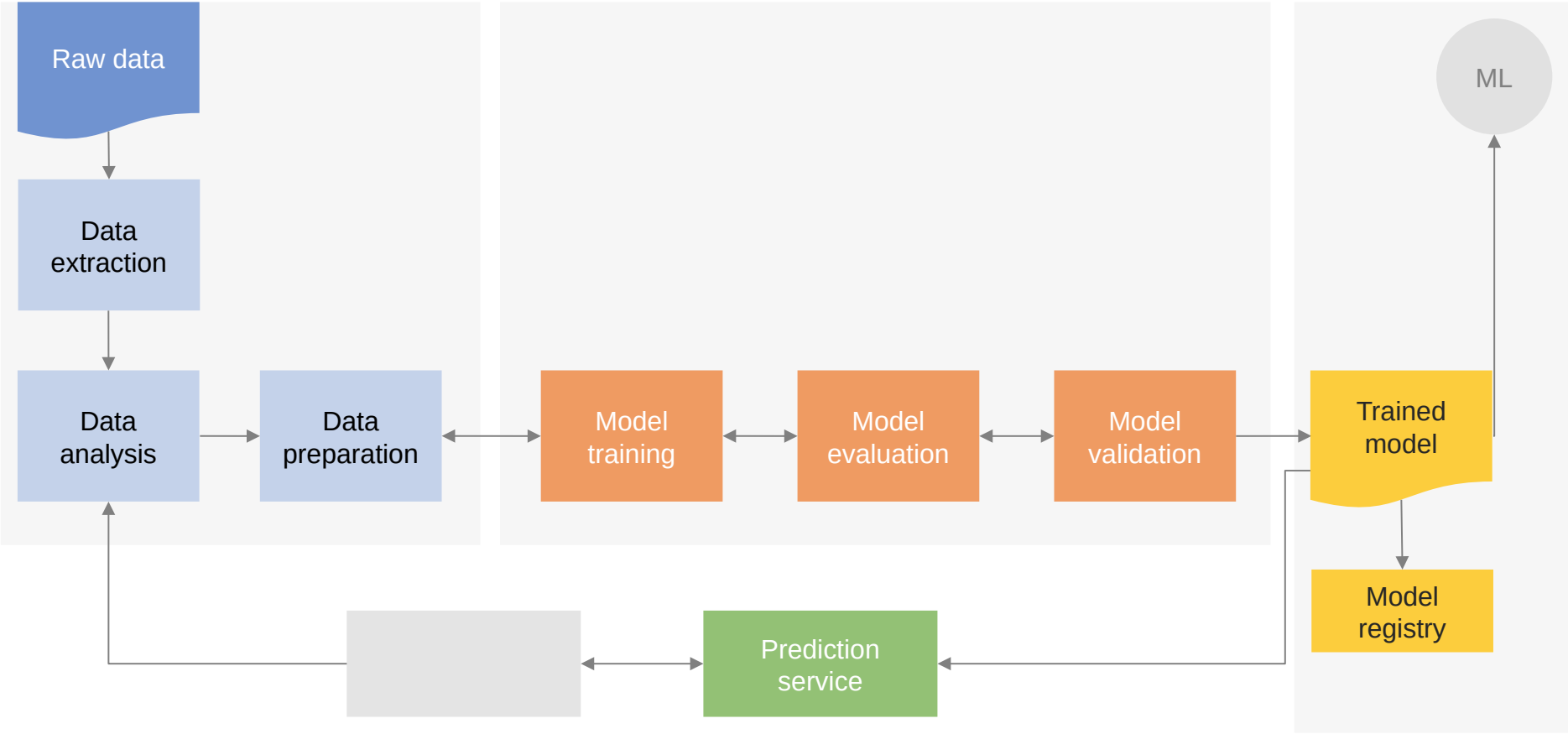

Raw data
Data extraction
Data analysis
Data preparation

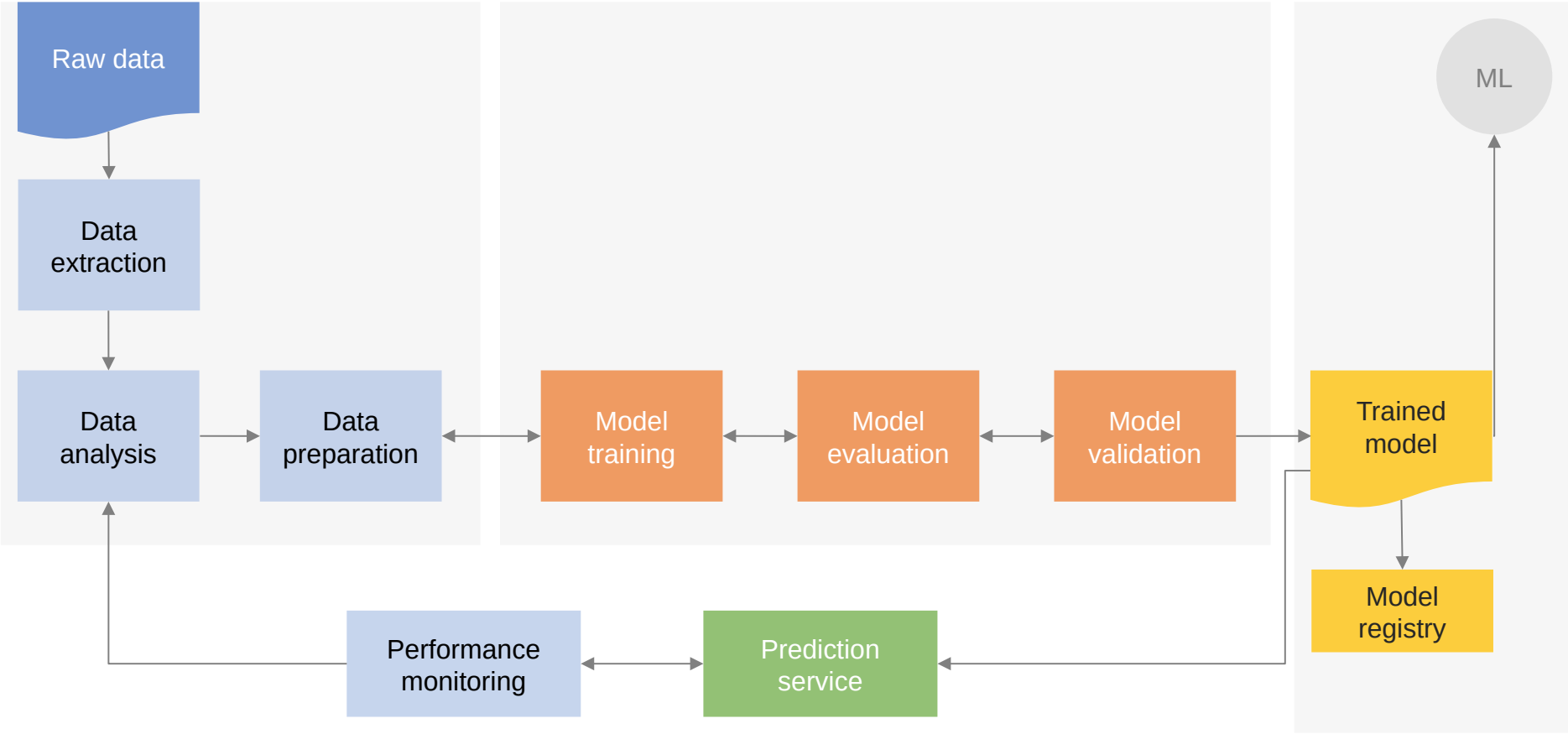

Raw data
Data extraction
Data analysis
Data preparation
Model training


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation

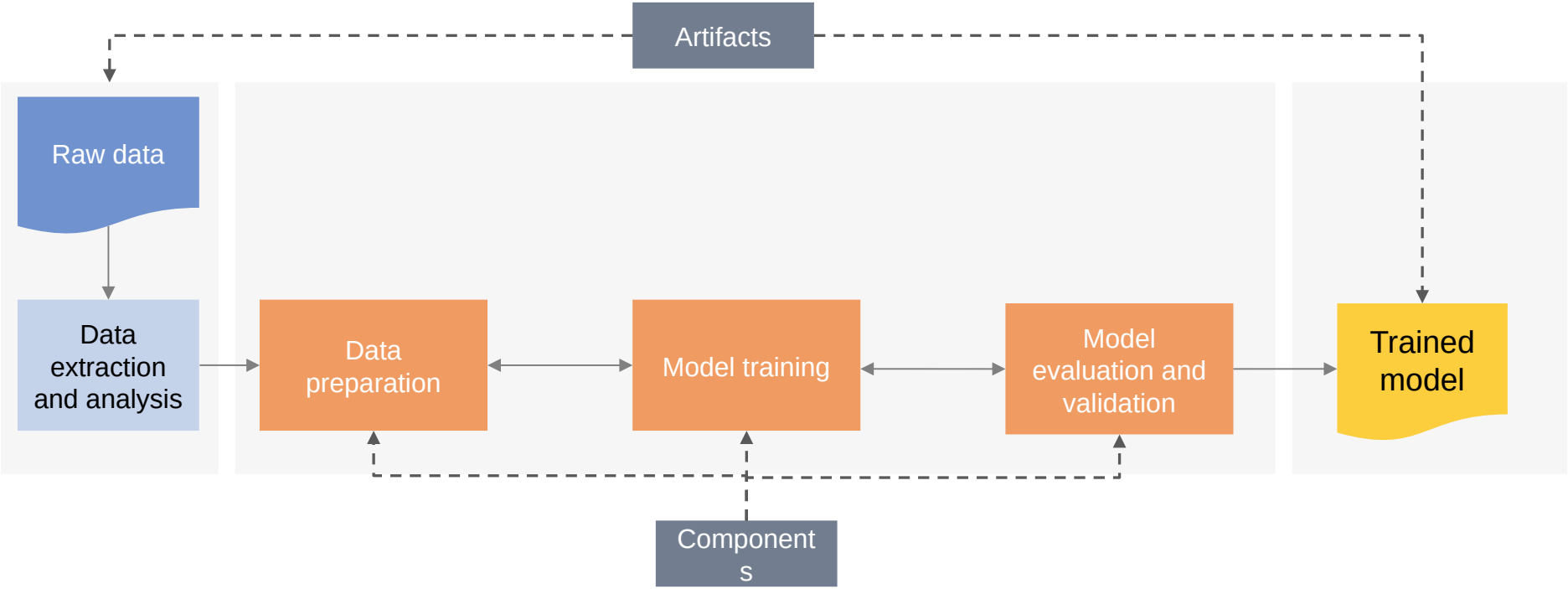

Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation
Trained model
ML
Model registry


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation
Trained model
ML
Model registry
Prediction service


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation
Trained model
ML
Model registry
Performance monitoring
Prediction service

ML Pipeline consists of artifacts and components



Components „do” things

Remember WBS?

Work Breakdown Structure

Model training

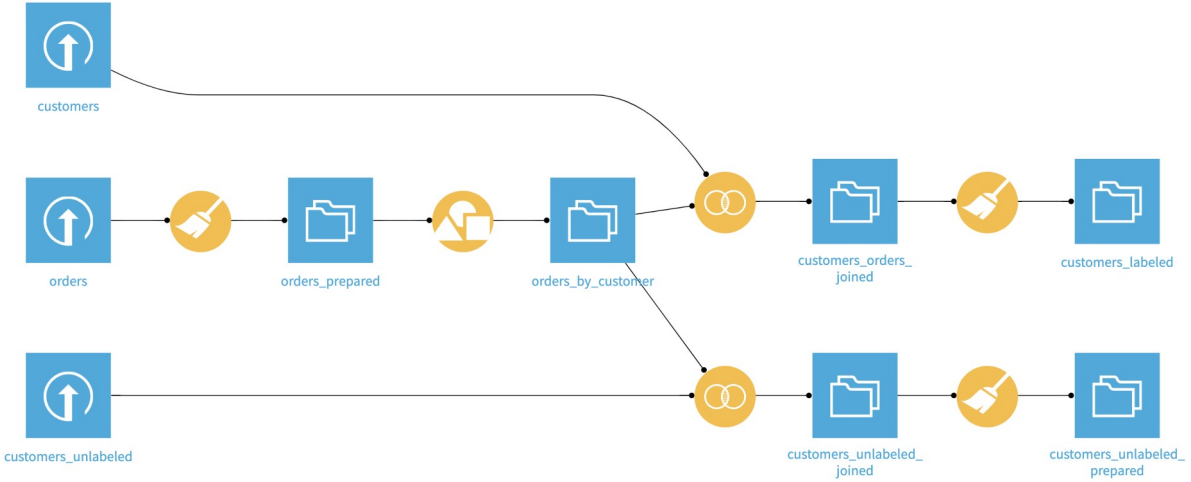
Artifacts are the „things”

Remember PBS?

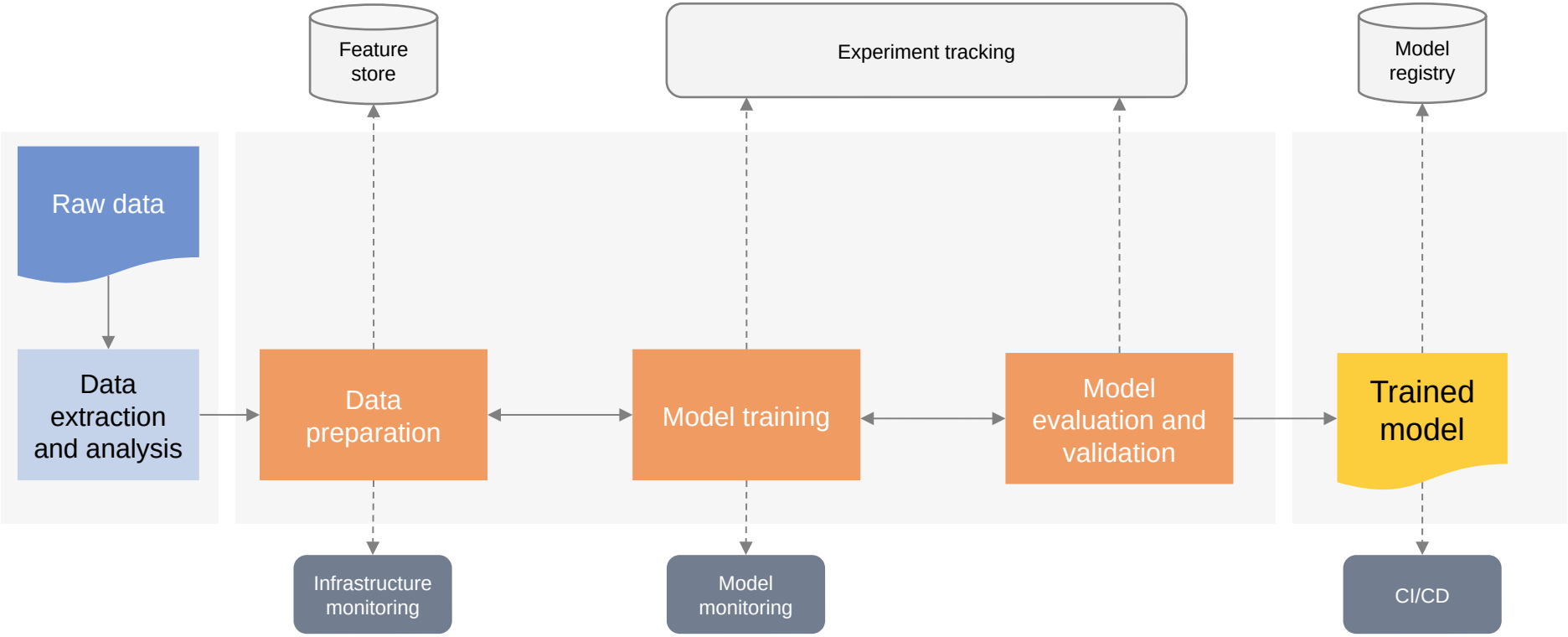
Product Breakdown Structure

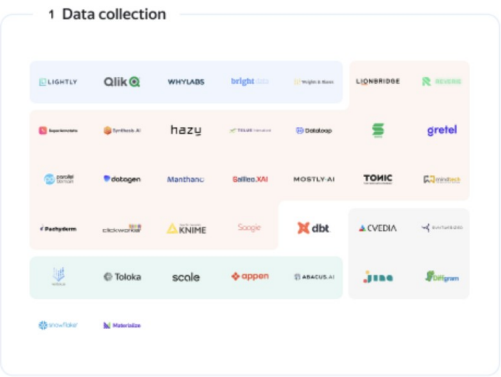
Trained model

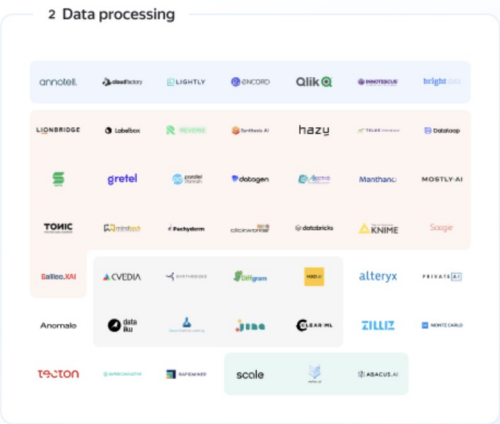
That’s often visualised in an ML pipeline

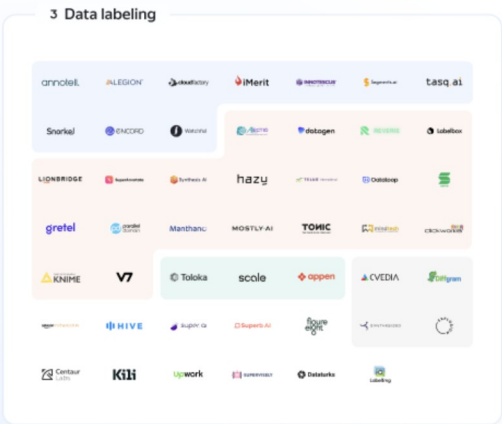


The whole proces is supported by repositories and CI/CD systems









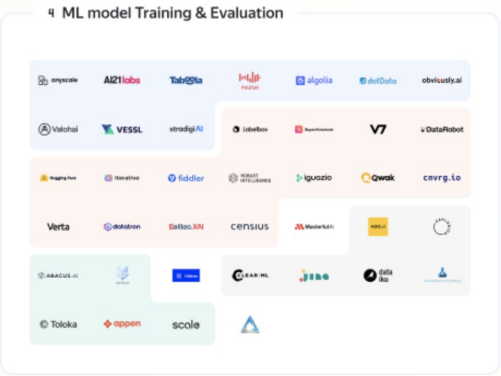
THE SEQUENCE

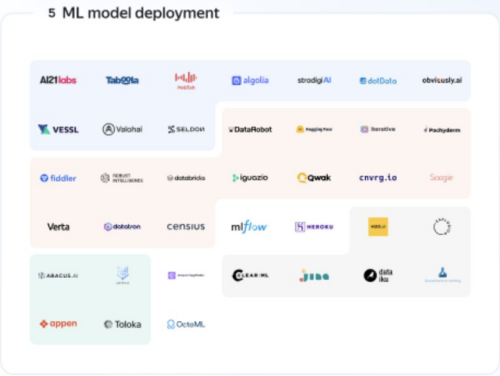
ML Value Chain Landscape

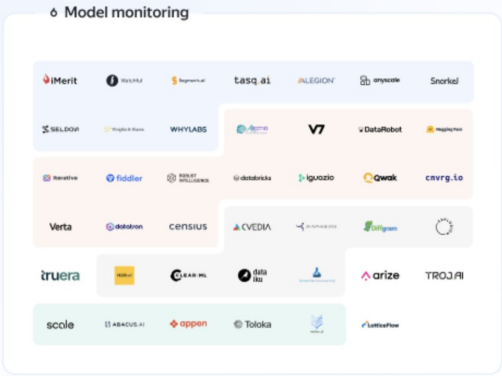
Created by your ML peers. Powered by TheSequence.

Vendors represented in more than 1 stage are indicated by color coded frames:

2 for the vendors covering **2** stages	3 for the vendors covering **3** stages	4 for the vendors covering **4** stages	5 for the vendors covering **5** stages







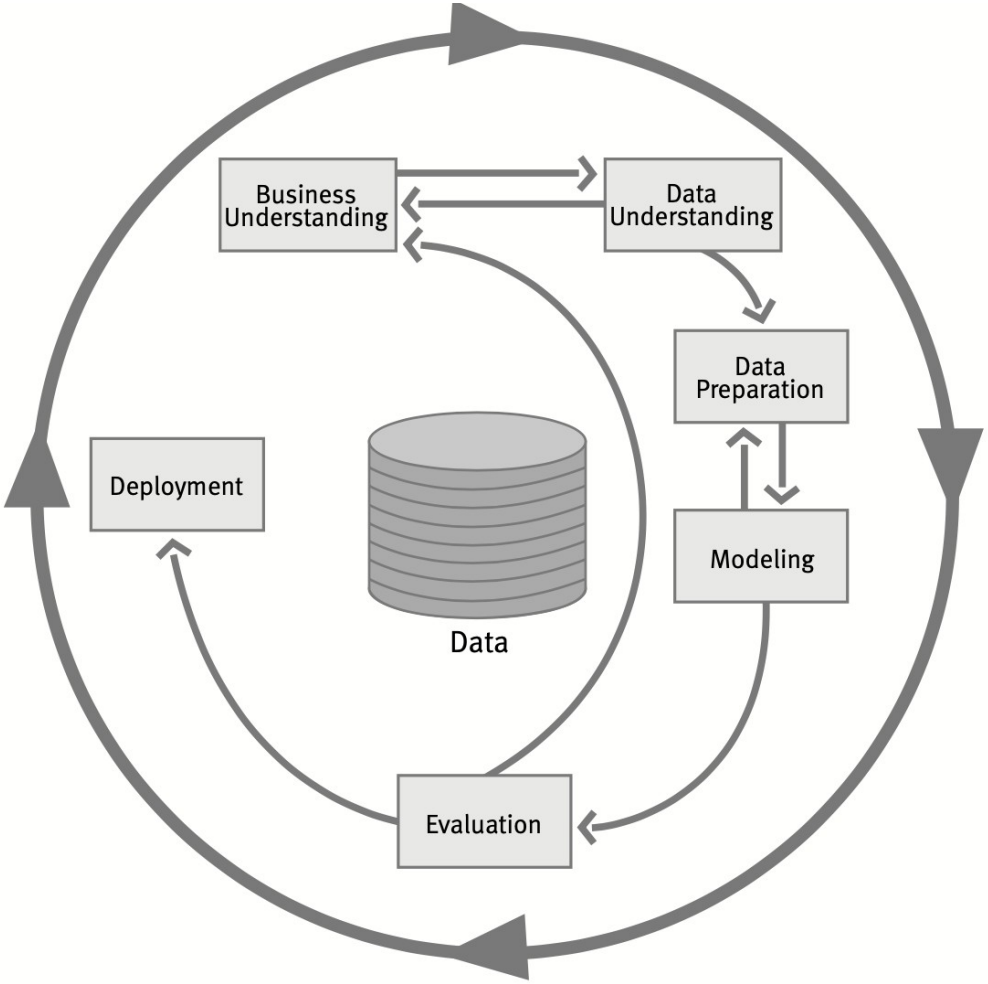
Dive In

Model Development

Develo
p
01
Deplo
y
02
Montio
r
03

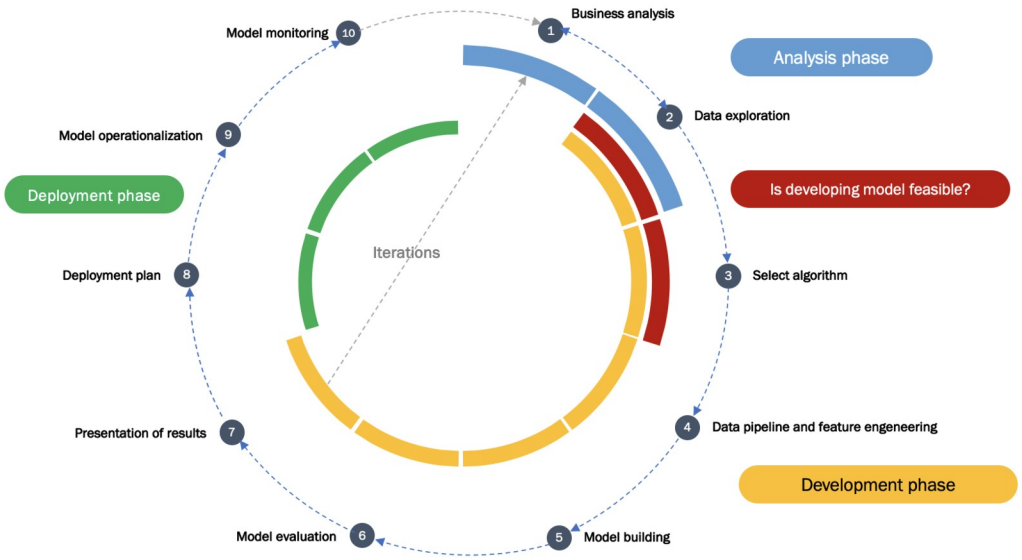
CRISP-DM

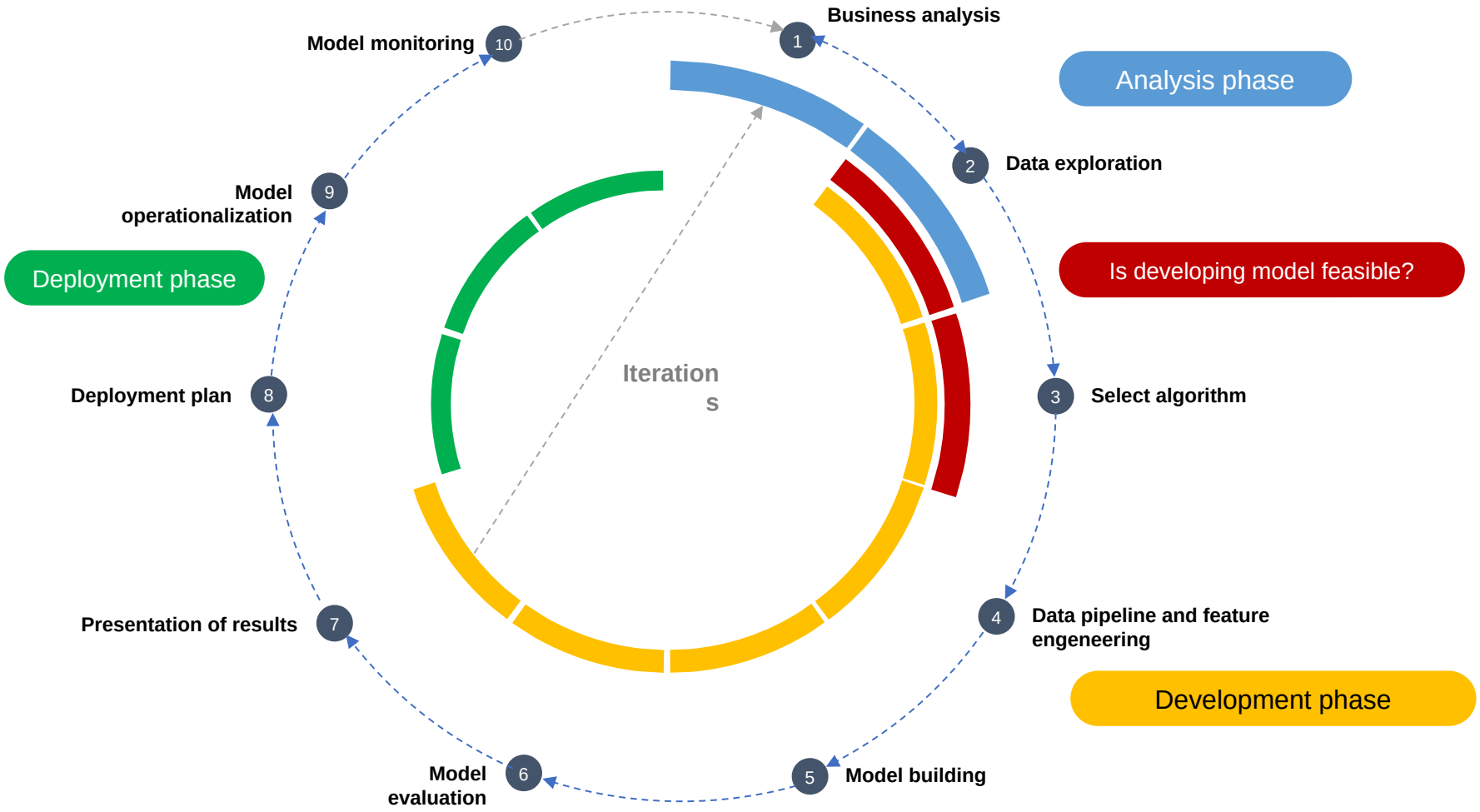
> 20 years old, but works ☺

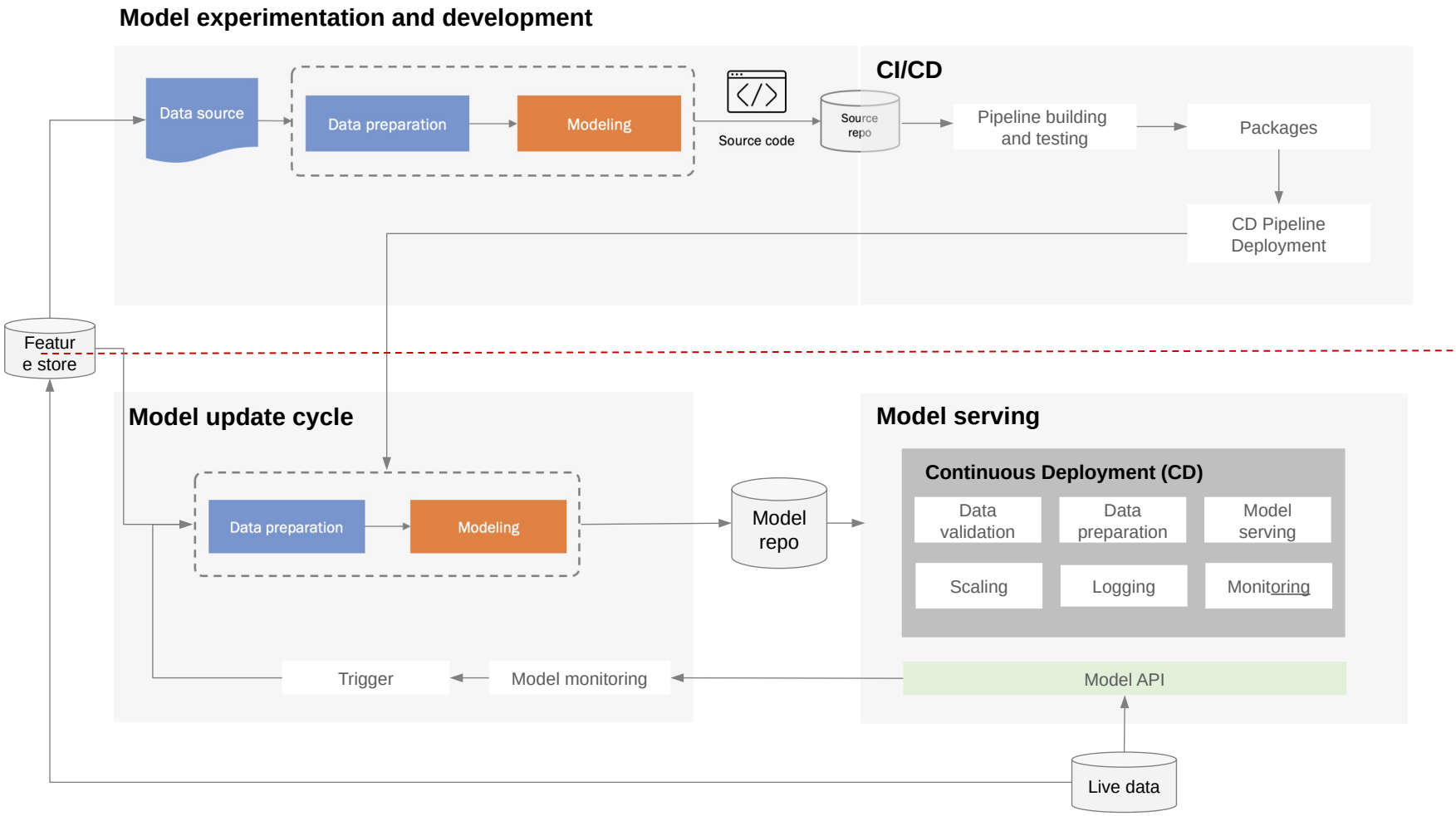


Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives *Background* *Business Objectives* *Business Success Criteria* **Assess Situation** *Inventory of Resources* *Requirements, Assumptions, and Constraints* *Risks and Contingencies* *Terminology* *Costs and Benefits* **Determine Data Mining Goals** *Data Mining Goals* *Data Mining Success Criteria* **Produce Project Plan** *Project Plan* *Initial Assessment of Tools and Techniques*	**Collect Initial Data** *Initial Data Collection Report* **Describe Data** *Data Description Report* **Explore Data** *Data Exploration Report* **Verify Data Quality** *Data Quality Report*	**Select Data** *Rationale for Inclusion/ Exclusion* **Clean Data** *Data Cleaning Report* **Construct Data** *Derived Attributes* *Generated Records* **Integrate Data** *Merged Data* **Format Data** *Reformatted Data* *Dataset* *Dataset Description*	**Select Modeling Techniques** *Modeling Technique* *Modeling Assumptions* **Generate Test Design** *Test Design* **Build Model** *Parameter Settings* *Models* *Model Descriptions* **Assess Model** *Model Assessment* *Revised Parameter Settings*	**Evaluate Results** *Assessment of Data Mining Results w.r.t. Business Success Criteria* *Approved Models* **Review Process** *Review of Process* **Determine Next Steps** *List of Possible Actions* *Decision*	**Plan Deployment** *Deployment Plan* **Plan Monitoring and Maintenance** *Monitoring and Maintenance Plan* **Produce Final Report** *Final Report* *Final Presentation* **Review Project** *Experience Documentation*

A bit more agile version




Business analysis
1
Analysis phase
2
Data exploration
Is developing model feasible?
3
Select algorithm
4
Data pipeline and feature engeneering
Development phase
5
Model building
6
Model evaluation
7
Presentation of results
8
Deployment plan
Deployment phase
9
Model operationalization
10
Model monitoring
Iteration s


Model experimentation and development
Data source
Data preparation
Modeling
Source code
Source repo
CI/CD
Pipeline building and testing
Packages
CD Pipeline Deployment
Featur e store
Model update cycle
Data preparation
Modeling
Model repo
Model serving
Continuous Deployment (CD)
Data validation
Data preparation
Model serving
Scaling
Logging
Monitoring
Trigger
Model monitoring
Model API
Live data